

Department of Telecommunications
Telecommunication Engineering Centre
(IoT Division)

F.No.-1-3/2022-IoT/TEC

Dated: 09.06.2023

Subject: Minutes of virtual meeting of National Working Group 20 (NWG-20) corresponding to ITU-T Study Group 20 on “Internet of Things (IoT) and Smart Cities & Communities (SC&C)” for the Study Period 2022-24, held on 5th June 2023, 2:30 PM.

The 10th virtual meeting of the National Working Group 20 (NWG-20) corresponding to ITU-T Study Group 20 was held on 5th June 2023 at 2:30 PM through remote participation using C-DOT VC platform. The list of participants is available as *Annexure-1*.

1. Mr. Sushil Kumar, DDG (IoT) welcomed all the participants in the meeting. While briefing about the agenda of the meeting, he mentioned that:-
 - a. In the last ITU-T SG-20 meeting held in Jan-Feb 2023, **Regional Group for Asia Pacific region (SG20RG-AP)** was created based on the Indian contribution and its chairmanship has been given to India. The first meeting of this Regional Group has been planned on 25-26 July 2023 in virtual mode. All the members having TIES account are requested to subscribe to its mailing list (details available on <https://www.itu.int/en/ITU-T/regionalgroups/sg20-ap/Pages/default.aspx>) for getting updates about this regional group and also to register for this meeting. Those not having TIES account and interested in participating and submitting contributions in these meetings, may intimate to TEC. Contributions submitted in this regional group will further be taken up in ITU-T SG-20 meeting. Also explained the structure of ITU-T SG-20 and its regional groups by presenting the webpage of SG-20.
 - b. IoT division, TEC has released nineteen Technical Reports (TRs) till date, covering various verticals, communication technologies and security, with the outcome intended to be used in the policy / standards. All the reports are available on TEC website (<https://www.tec.gov.in/M2M-IoT-technical-reports>).
 - c. International Telecommunication Union (ITU) has posted the following six TEC Technical Reports on its website (<https://www.itu.int/cities/dt-resource-hub/iot/>) in IoT sections (2023, 2022 and 2021), recognizing as insightful technical resource for the benefit of global community-
 - i. Security by design for IoT Device Manufacturers

- ii. Framework of National Trust Centre for M2M/IoT Devices and Applications
 - iii. IoT/ ICT Standards for Smart Cities
 - iv. Emerging Communication Technologies & Use Cases in IoT Domain
 - v. Code of Practice for Securing Consumer Internet of Things (IoT)
 - vi. IoT/ ICT Enablement in Smart Village and Agriculture
- d. Agenda of this meeting is to discuss how to increase participation in the meeting of ITU-T SG-20 Regional Group for Asia Pacific and prepare contributions for the same.
- e. Following three contributions submitted by IoT division TEC have been approved in **ITU/ FAO Focus Group on 'Artificial Intelligence (AI) and Internet of Things (IoT) for Digital Agriculture' (FG-AI4A)** meetings in 2022-23-
- i. *Applications of Drones, AI and IoT in Cashewnuts farming (based on the project in VIT Chennai)*
 - ii. *IoT based Farmland Surveillance System with Disease Detection in Paddy Crops (based on the project in VIT Chennai)*
 - iii. *Artificial Intelligence-based Disease Identification in Wheat Crops (based on the project in ICAR, New Delhi)*

These were presented jointly by TEC officers and the members from VIT/ ICAR.

- f. The ITU webpage link for accessing SG-20 documents is <https://www.itu.int/md/T22-SG20-230130-TD/en>. Members may visit this webpage and send the document number, title and related question of the document(s) of their interest to TEC, on which they wish to work. These documents will be downloaded and shared with the concerned members. Contributions may either be submitted as inputs to the existing on-going work items or by proposing creation of a new work item, on which no previous related work is available.
- g. The drafts of six on-going work items in ITU were circulated by ADG (IoT) via e-mail to all the NWG-20 members along with minutes of the previous meeting. He also presented all the work items in brief.
2. Mr. Sharad Arora, Mashmari presented on the newly created work item by him in ITU-T SG-20- Y.RemoteEd *"Requirements, capabilities and architectural frameworks for e-learning in remote classrooms"*. Mr. Prateep Misra, TCS, Dr. Ekram Khan, AMU and Mr. Sachin Chaudhari, IIIT-H expressed their willingness to contribute in this work item. DDG (IoT) requested Mr. Sharad to create a group of the interested members to further carry out the work on this work item.

3. Mr. Kishor Narang, Narnix, volunteered to submit contributions on Unified digital infrastructure for Smart Cities, Unified Last Mile Information Protocol Stack Reference Architecture for IoT and Smart City use cases and IEC Smart City Reference Architecture project. DDG (IoT) suggested to discuss with him in one-to-one meeting.
4. Dr. R Lenin Raja, AAEMT labs agreed to prepare one contribution on the EMF exposure from IoT devices, based on the globally available standards.
5. Mr. Pranav Singh, Idemia expressed his willingness to submit contributions in Q6 (Security, privacy, trust and identification for IoT and SC&C).
6. Mr. Sudipta Ghosh, Wipro mentioned to work on IoT security and privacy aspects.
7. DDG (IoT) summarized by mentioning that we may work on some on-going work items already shared with the members. He requested members to send their contributions on priority for discussion in the next meeting so that the same may be submitted for the upcoming regional group meeting.

In the end, DDG (IoT) thanked all the participants for attending the meeting.

Way Forwards:

1. Next meeting of NWG-20 will be held on **19th June 2023 at 2:30 PM.**
2. All members having TIES account are requested to subscribe to the mailing list of ITU-T SG20 Regional Group for Asia Pacific region (SG20RG-AP) and also to register for its meeting scheduled on 25-26 July 2023 (details available on <https://www.itu.int/en/ITU-T/regionalgroups/sg20-ap/Pages/default.aspx>).
3. One-to-one meetings will be held in-between with the members for discussing their contributions.

This is issued with the consent of DDG (IoT).

Encl: As above

Hindi version shall follow.


(Namrata Singh)
ADG (IoT)
TEC, New Delhi

Annexure-1

S. No.	Name	Organisation
1.	Sushil Kumar	TEC
2.	Ms. Ashima	TEC
3.	Ms. Namrata Singh	TEC
4.	Rajneesh Kumar	TEC
5.	Shekhar Singh	TEC
6.	Ms. Jyoti Sengar	TEC
7.	Ms. Riya Soy	TEC
8.	Alok Mittal	STMicroelectronics
9.	Manoj Desai	ARAI
10.	Ankur Chakraborty	TCS
11.	Prateep Misra	TCS
12.	Dr. R Lenin Raja	AAEMT labs
13.	Prof. Ekram Khan	AMU
14.	Hrushikesh	Airtel
15.	Parag Kulkarni	JohnDeere
16.	Kishor Narang	Narnix
17.	Venkata Praveen Pogiri	Director WF, DoT
18.	Naveen Khichar	DoT HQ
19.	Neelesh Mantri	BP
20.	Rajith Ali	SIA India
21.	Sachin Chaudhari	IIIT-H
22.	Sharad Arora	Mashmari
23.	Sudipta Ghosh	Wipro